PROJECT

Price Optimization for E-commerce Products Using Regression Models

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Predictive Analytics

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Phase I: Project Proposal

A. Project:

Price Optimization for Ecommerce Products Using Regression Models.

B. Introduction:

Price is very important for the success of businesses. If the price is too high people might not buy. If the price is too low the business might not make money. That is why businesses need to find the balance between getting customers selling more and making money.

This project will look at how regression models can help online businesses decide prices. By looking at sales data the project will study how product prices relate to sales. Different regression models will be made to predict sales using things like product prices, deals, product categories and other things in the data. These predictions will then be used to compare prices and find prices that might help make more money.

The value of this project is that it uses data and machine learning to help make business choices. Of just using experience or guesswork businesses can use old data to understand how customers buy things and test different prices. These findings can help make money help plan inventory and make the company better in the competitive online world.

Research Questions:

1. How do product prices and sales performance connect in the data?

2. How well can regression models predict sales using price and other things about the product?

3. Which regression model gives the sales predictions?

4. How can predicted sales be used to compare prices and find ways to make more money?

Project Objectives:

- Look at old online sales data to see how product prices and sales are related.
- Find the things that might affect how much a product sells.
- Make and compare different regression models to predict sales.
- Check how well the models work using the right measures.
- Try different product prices and see what they could do to sales and money.
- Give advice on prices based on the results.

C. Methodology:

The project will follow a step-by-step process to look at data starting with looking at the data and ending with checking models and giving price advice.

1. Data Understanding and Exploratory Data Analysis

The first step is to look at the data and understand what it has. Statistics and pictures will be used to check prices, sales, deals and money based on what's available. Histograms, scatter plots and looking at how things connect will help find patterns and see how pricing might link to sales. This step will also find values and decide which things are useful for the analysis.

2. Data Preprocessing

Before making models, the data will be cleaned up. Ready for analysis. This will include getting rid of the records handling missing data, fixing data types and checking for bad or extreme values. Things like product categories will be changed into a format that works when needed. Numbers might be scaled if the models need it. The main thing to predict and the things to use for that will be picked based on what's in the data.

3. Feature Selection and Engineering

Next the things that might help explain or predict sales will be found. Product prices will be a thing to look at along with other things that might affect buying. For example, if the data has prices and how many are sold those could be used to look at demand. Whether money or the value of sales is in the data that might be looked at by itself. Deals and product types will be used if they are in the data and important to the goals.

4. Regression Model Development

Several models will be. Compared to see which one predicts sales best. Linear Regression will be the starting point because it shows how things are connected clearly. Then a better model like Classification and Regression Trees or Random Forest Regression will be used to see if it can find complex links between product features and sales. The data will be split into training and testing parts with 80% for training and 20% for testing. The training data will build the models and the testing data will check how well they work on data.

5. Price Optimization

After checking the models, the best one will be used to test prices for some products. A range of prices will be tried while keeping things the same. The model will guess how many will be sold at each price.

The revenue made from each price will then be calculated using this formula:

Revenue = Price × Predicted Number Sold

The price that gives the money in the tested range will be the one that makes the most money according to the model.

If there is information about the cost to make the product the project will also look at how much money's made using this formula:

Profit = (Price – Unit Cost) × Predicted Number Sold

This extra check is important because the price that makes the most money might not be the same as the one that makes the most profit.

D. Dataset:

The project will use the Ecommerce Sales Data dataset from Kaggle.

Dataset Source: [Kaggle – Unlock Profits with Ecommerce Sales Data] (https://www.kaggle.com/datasets/thedevastator/unlock-profits-with-e-commerce-sales-data)

This data will be used to look at how products sell and see what links there are between prices and how much they sell. Before starting the analysis, the datas description and the file will be checked to make sure there are records, enough columns, the right things and that the data is good for the project goals.

The size and way the data is set up will be shared after looking at the file not from a count.

Planned Steps to Clean the Data:

- Look at the data. Find the right things to use.
- Get rid of the same records when needed.
- Fix wrong values.
- Change prices, numbers, deals and money into numbers.
- Look for values and check for any that might be wrong.
- Change things that are words into numbers when needed.
- Pick what to predict and what to use for that based on the data.
- Split the data into parts for training and testing.

One problem could be if the data does not have different prices or sales to make good price analysis. If the cost to make the product is not there the project will focus on making the money instead of saying anything about profit.

Also, if there are not similar products or things that are priced in different ways the results will be seen as ideas to think about. Since the data is from the past the analysis might find links between price and sales. It can't say for sure that changing the price will change how much is sold.

E. Evaluation:

The models will be checked using three measures then the money made from different prices will be compared.

1. Mean Absolute Error (MAE)

MAE shows the difference between real sales and the estimated ones. A smaller MAE means the models guesses are closer to the numbers.

2. Root Mean Squared Error (RMSE)

RMSE looks at the errors. Gives more weight to bigger mistakes. A smaller RMSE usually means the model works better especially when big mistakes matter a lot for the business.

3. Coefficient of Determination (R^2)

R^2 shows how much of the changes in the thing being predicted are explained by the model. A higher R^2 means the model explains more of what's seen.. It will be checked along with MAE and RMSE because a high R^2 does not always mean the model guesses right every time.

4. Money Comparison

The final check will look at the money made from prices. The number of sales the model predicts will be used to calculate money at the price and the new prices being tested.

The change in money made will also be calculated to show how better the model thinks things could be. These results will be shown as guesses from the model not as things in real business situations.

The model that works best on the test data will be used for the price changes. Final advice will also look at whether the sales numbers make sense if the new prices are, in a range that makes sense with the data and if the old data is enough to back up the findings.

References

Devastator, T. (n.d.). *Unlock Profits with E-commerce Sales Data*. Retrieved from Kaggle: https://www.kaggle.com/datasets/thedevastator/unlock-profits-with-e-commerce-sales-data

James, G. (2023). An Introduction to Statistical Learning: With Applications in Python. New York: Springer.

scikit-learn developers. (n.d.). *Linear Regression*. Retrieved from scikit-learn documentation: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html

scikit-learn developers. (n.d.). *Random Forest Regression*. Retrieved from scikit-learn documentation: https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html

scikit-learn developers. (n.d.). *Regression Metrics*. Retrieved from scikit-learn documentation: https://scikit-learn.org/stable/modules/model_evaluation.html#regression-metrics